

Chapter 34: The Preterm Newborn — Revision Notes

Williams Obstetrics 26e, pp. 1588–1611 of the PDF. High-yield numbers are in **bold**. The one table is transcribed from the PDF image (it is missing from the extracted chapter text). All 3 figures are included from the PDF.

Big picture: Viability is now **22–24 wk**, and survival for neonates **<29 wk approaches 80%**. Still, **two thirds of US infant deaths** occur in the **10%** born before 37 wk. The main threats are RDS/BPD, NEC, ROP and brain injury (IVH, PVL → CP).

1. Overview and Complications of Prematurity

Table 34-1. Complications of Prematurity

Complication
Respiratory distress syndrome (RDS)
Bronchopulmonary dysplasia (BPD)
Pneumothorax
Pneumonia/sepsis
Patent ductus arteriosus (PDA)
Necrotizing enterocolitis (NEC)
Retinopathy of prematurity (ROP)
Intraventricular hemorrhage (IVH)
Periventricular leukomalacia (PVL)
Cerebral palsy (CP)
Neurodevelopmental impairment (NDI)

- US preterm birth rate fell from **~12% (2007) → 10% (2020)**.
- Early preterm (<34 wk): **2.77% → 2.69%**, the lowest since 2007.

2. Respiratory Distress Syndrome

- **The seminal complication of prematurity:** immature lungs + **surfactant deficiency** → hypoxia.
- Hypoxia → pulmonary hypertension and neurological damage (e.g., CP).
- **Hyperoxia** from treatment contributes to **BPD, NEC, PVL and ROP**.

Etiopathogenesis

- At birth, lungs must fill with air and clear fluid while pulmonary blood flow rises. Some fluid is squeezed out during vaginal birth; **most is absorbed by pulmonary lymphatics**.
- **Surfactant** (from **type II pneumocytes**) lowers surface tension → prevents alveolar collapse at end-expiration.
- Can occur at term with **sepsis, pneumonia or meconium aspiration** (surfactant inactivated by inflammation or meconium).
- Sequence without surfactant:
 - Unstable alveoli collapse → hypoxia and hypotension impair pneumocyte nutrition.
 - Partial persistence of fetal circulation → **pulmonary hypertension and right-to-left shunt**.
 - Alveolar cells undergo ischemic necrosis. With O₂ therapy the pulmonary bed dilates and the shunt reverses; protein-rich fluid leaks into alveolar ducts and lining cells slough.

- **Hyaline membranes** (fibrin-rich protein + cell debris) line dilated alveoli and terminal bronchioles; underlying epithelium necroses.
- On H&E the membranes are **amorphous and eosinophilic, like hyaline cartilage** → hence "**hyaline membrane disease.**"

Clinical course

- **Tachypnea, chest wall retraction, nostril flaring, expiratory grunting** (grunting creates auto-PEEP to prevent collapse).
- Shunting through unventilated lung → hypoxemia, **metabolic and respiratory acidosis**; poor peripheral perfusion, hypotension.
- **Chest radiograph: diffuse reticulogranular infiltrate + air bronchograms.**
- Differential of respiratory insufficiency: sepsis, pneumonia, meconium aspiration, pneumothorax, persistent fetal circulation, heart failure, thoracic malformations (e.g., **diaphragmatic hernia**). Rare: surfactant protein or **ABCA3** transporter mutations.

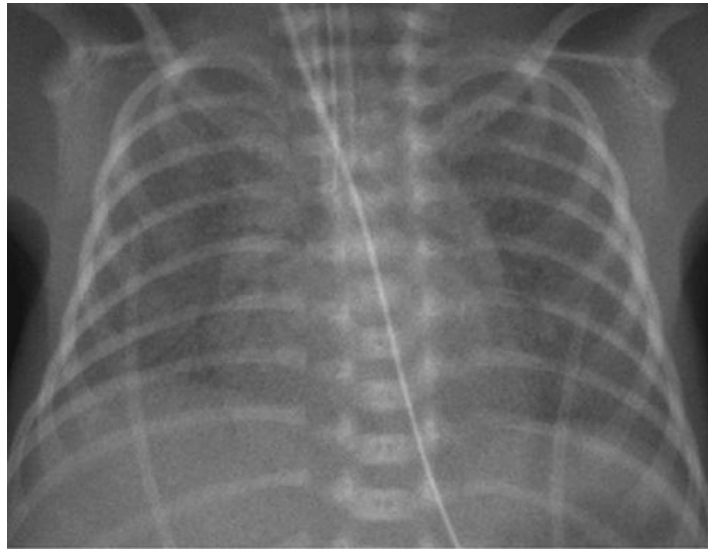


Figure 34-1. Moderate RDS: diffuse granular ("ground-glass") lungs with central air bronchograms; only 8 ribs expanded despite ventilation.

Treatment

Ventilatory strategy

- Supplemental O₂ is needed, but **excess O₂ damages lung epithelium, retina and other immature tissues.**
- **CPAP** keeps unstable alveoli open → allows lower FiO₂ (less toxicity).
- **Initial CPAP + selective surfactant** is a beneficial alternative to immediate intubation + surfactant in extremely preterm neonates.
- **Less-invasive surfactant administration (thin catheter on CPAP)** → ↓ intubation/ventilation in the first **72 h**, ↓ **death or BPD**, ↓ major complications and in-hospital mortality vs ET tube delivery.
- High-frequency oscillatory ventilation: variable benefit/risk. High-frequency jet ventilation: not superior to conventional ventilation.

Bronchopulmonary dysplasia (chronic lung disease of prematurity)

- **The most frequent complication of extreme preterm birth.**
- Causes: mechanical ventilation (**barotrauma, volutrauma**), **hyperoxia** (reactive oxygen species → inflammation), infection.
- Disrupted alveolar and pulmonary vascular development → hypoxia, hypercarbia, **chronic O₂ dependence**; can affect adult lung function.

BPD prevention/treatment	Verdict
Postnatal corticosteroids (e.g., dexamethasone)	AAP recommends against routine use — limited benefit, ↑ impaired motor/cognitive function and school performance. Worst when given to infants at low BPD risk. Intratracheal steroid + surfactant under study.
Inhaled nitric oxide	NIH consensus: data do not support use to prevent or treat BPD
Caffeine (for apnea of prematurity; bronchodilator)	↓ BPD, better early neurodevelopment, safe to 11 years → widely used at ≤1250 g
Vitamin A (antioxidant; preterm infants have low levels)	Modestly ↓ BPD in VLBW <1500 g ; limited by production shortages

Surfactant prophylaxis and rescue

- **Standard therapy for RDS:** ↓ mortality, ↓ need for ventilation, better short-term outcomes.
- Usually given by **endotracheal tube**. Animal-derived products: **bovine (Survanta), calf (Infasurf), porcine (Curosurf)**.
- **Animal-derived > synthetic** (synthetic lacks key surfactant proteins); **no synthetic surfactant is currently available**.
- Less invasive rescue routes: pharyngeal instillation, nebulization, laryngeal mask, **thin catheter on CPAP** (best evidence).
- Antenatal corticosteroids + postnatal surfactant → an even greater ↓ in death.
- ⚠️ **Prophylactic surfactant is no longer beneficial** where antenatal steroid use is high and delivery-room CPAP is routine — it is linked to ↑ **death or BPD**.

Prevention

Antenatal corticosteroids

- **NIH 2000:** a single course ↓ **RDS and IVH** in neonates born **24–34 wk**.
- **ACOG:** all women at risk of preterm birth in this range are candidates; may be considered from **23 wk** if delivery is expected **within 7 days**.
- **Perivable period:** tie steroid use to the family's **resuscitation decision** (see Table 45-4).
- **Late preterm (34–36 wk):** ↓ neonatal respiratory complications; ACOG supports considering a **single course of betamethasone**. **Parkland does not currently do this**.

Amniocentesis for fetal lung maturity

- Previously used to predict RDS. **No longer supported or recommended** to guide delivery in any clinical situation.

3. Necrotizing Enterocolitis

- Findings: **abdominal distension, emesis, ileus, bilious gastric aspirates, bloody stools**.
- **Radiology: pneumatosis intestinalis** (intramural gas from invading bacteria) — classic; also **hepatobiliary/portal venous gas** and **pneumoperitoneum**.
- Mainly **low-birthweight** newborns; occasionally mature ones.
- **Hypothesized causes:** perinatal hypotension, hypoxia, sepsis, **umbilical catheterization**, exchange or blood transfusions, **cow milk and hypertonic feeds**.
- **Pathophysiology is multifactorial:** genetics, intestinal immaturity, microvascular tone imbalance, **abnormal microbial colonization**, enteral feeds, highly immunoreactive mucosa.

Management	Details
Medical	Abdominal decompression, bowel rest , broad-spectrum antibiotics , parenteral nutrition
Surgical	For perforation or clinical/biochemical deterioration: drain placement, laparotomy with resection, or enterostomy (stoma). Perforation → prompt resection .

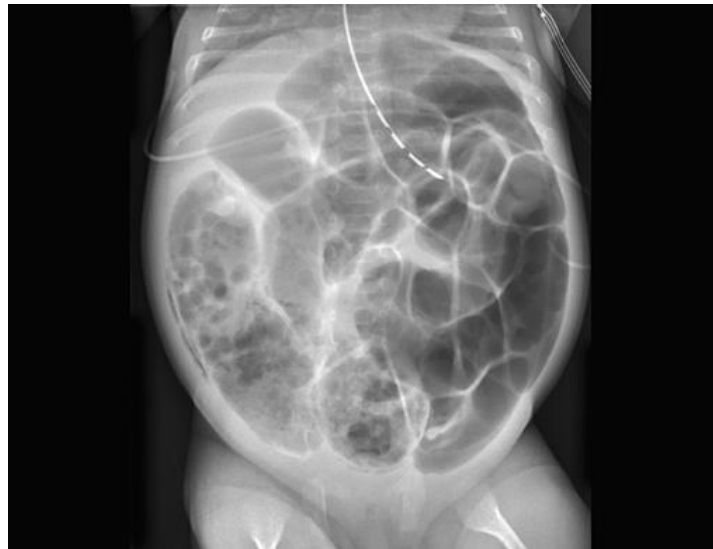


Figure 34-2. NEC: distended gas-filled small and large bowel with pneumatosis intestinalis (intramural gas) in the right lower quadrant and pelvis.

4. Retinopathy of Prematurity

- Formerly **retrolental fibroplasia**; by **1950 the largest single cause of blindness** in the US. Fell once linked to **hyperoxemia**, then rose again as more extremely preterm infants survived.
- **Normal**: retina vascularizes **centrifugally from the optic nerve** from **~4th month** until shortly after birth.
- **Pathogenesis**: excess O_2 → severe retinal **vasoconstriction** → endothelial damage and vessel obliteration → **aberrant neovascularization** into the vitreous → leakage, bleeding → **adhesions and retinal detachment**.
- **VEGF** is upregulated in ROP → basis for **anti-VEGF therapy**.
- Even room air after birth is "relatively" hyperoxic compared with in utero. The safe O_2 level is unknown.

SUPPORT trial (1316 neonates, 24–27 wk)

SpO ₂ target	Death before discharge	Severe ROP in survivors
85–89% (lower)	20% (significantly ↑)	8.6% (significantly ↓)
91–95% (higher)	16%	17.9%

Memory hook: "Low O_2 saves the **eyes** but costs **lives**."

5. Brain Disorders

- Preterm lesions differ from term: **IVH**, **cerebellar hemorrhage**, **periventricular hemorrhagic infarction**, **cystic PVL**, **diffuse white-matter injury** — all strongly linked to adverse neurodevelopment.
- **Cranial sonography is the preferred modality** in preterm neonates (available, reliable, tracks brain growth).
- Cystic injury takes **2–5 wk** to evolve → **serial scans**. Transient lesions that resolve carry a better prognosis.
- **4–10%** of preterm children develop CP **without** lesions; i.e., **90–96%** of preterm newborns with CP have lesions visible on cranial sonography.

Intracranial hemorrhage — five categories (Table 33-4)

Type	Typical population	Significance
Primary subarachnoid	More common preterm	Frequently benign
Cerebellar	More common preterm	Increasingly recognized cause of serious sequelae
Intraventricular (IVH)	Almost exclusively preterm ; relatively common	Can be serious
Subdural	More common term	Can be serious
Miscellaneous intraparenchymal	More common term	Variable

Intraventricular hemorrhage

Pathogenesis

- **Germinal matrix** = highly cellular, vascular region **beneath the lateral ventricles**. Its capillaries are fragile because of:
 - (1) poor support from the subependymal matrix
 - (2) venous anatomy causing **stasis and congestion** (vessels burst if pressure rises)
 - (3) **impaired autoregulation**
- Rupture → blood in ventricles → ventricular dilation → obstructed venous drainage → **periventricular hemorrhagic infarction**.
- Risk of severe IVH is **inversely related to GA**: serious **<32 wk**, especially frequent **<27 wk**.
- **Germinal matrix involutes at 34–36 wk** → IVH infrequent after that.
- **Most bleed within 72 h** of birth — so often wrongly blamed on birth events; **prelabor IVH also occurs**.
- Multifactorial: hypoxic-ischemic events, **↑ CO₂**, anatomy, **BP instability**, coagulopathy, genetics; **infection** (endothelial activation, platelet adherence, thrombi); RDS and mechanical ventilation.
- **~Half are clinically silent**. Most small germinal matrix/intraventricular bleeds resolve, yet nearly half show some neurological impairment. Extensive bleeds → major handicap, **hydrocephalus** or **PVL**. **PVL extent correlates with CP risk**.

Incidence

- NRN, <29 wk: **19% (1993) → 13% (2012)**.
- 2012: **11% at 23 wk → 5% at 28 wk**.

Papile grading (modified)

Grade	Extent
I	Limited to the germinal matrix
II	IVH occupying 10–50% of the ventricular area
III	IVH with >50% of the ventricular area
Parenchymal hemorrhage (formerly grade IV)	Most likely hemorrhagic venous infarction , not IVH extending into parenchyma

Antenatal corticosteroids and IVH

- Given **≥48 h before delivery** → prevent or ↓ severe IVH.
- **NIH 1994**: ↓ mortality, RDS and IVH at **24–32 wk**. **NIH 2000**: **no repeated courses**.
- **MFMU (Wapner) repeat courses**: some better neonatal outcomes but **↓ birthweight and ↑ FGR**; no difference in physical/neurocognitive measures at 2–3 yr, but a nonsignificant **5.7-fold RR of CP**.
- **Australian trial (>1100)**: CP **4.2% vs 4.8%** (repeat vs single) — about the same.
- <28 wk: **≥10 days** since betamethasone → ↑ severe IVH (inconsistent across studies).

- **ACOG now:** a **single course at 24 0/7–33 6/7 wk** for women at risk of preterm delivery; a **second "rescue" course** if the first was **>14 days** earlier and delivery is imminent.

Other preventive methods

Intervention	Effect
Antenatal magnesium sulfate	Does not ↓ IVH, but protects against neurodevelopmental impairment → ACOG recommends
Antenatal vitamin K or phenobarbital; postnatal phenobarbital	No consistent ↓ in IVH
Vitamin E	↓ IVH but ↑ sepsis
Postnatal indomethacin / prostaglandin inhibitors	Inconsistent ↓ IVH; no neurodevelopmental benefit
Cesarean (VLBW)	No effect on severe IVH; ↓ overall IVH (one meta-analysis). Unclear for breech/FGR.
Delayed cord clamping	↓ any-grade IVH (not severe IVH); ↑ Hb/Hct; may ↑ survival
⚠ Cord milking	↑ severe IVH at <28 wk → avoid

Periventricular leukomalacia

- **Cystic areas deep in white matter** after hemorrhagic or ischemic infarction → regional necrosis.
- No regeneration and minimal gliosis in preterm brain → **echolucent cysts** on imaging.
- Take **≥2 wk** to form (up to **4 months**) → **cysts present at birth imply an earlier (antenatal) insult**, helping to time the event.

6. Cerebral Palsy

- **Definition:** chronic movement or posture abnormalities, **cerebral in origin**, arising early in life, **nonprogressive**. Epilepsy and intellectual disability often coexist.
- Classified by dysfunction (spastic, dyskinetic, ataxic) and limb distribution (quadriplegia, diplegia, hemiplegia, monoplegia).

Type	Features
Spastic quadriplegia	Most severe spastic form: all four limbs, trunk, face; strong link to developmental delay and epilepsy
Spastic diplegia	Common in preterm/LBW ; stiffness mainly in legs (arms less or not affected); legs pull together, turn in, cross at the knees → walking difficulty
Spastic hemiplegia	One side only; arm > leg
Dyskinetic	Athetoid, choreoathetoid, dystonic; slow writhing or rapid jerky movements; hard to sit and walk
Ataxic	Balance and coordination problems
Mixed	>1 type; commonest is spastic–dyskinetic

Incidence

- US prevalence **~3 per 1000 children**.
- Sweden, born **<28 wk** (2004–2007), 6-yr follow-up: **CP 10.5%**, with **no linear association** with GA in this cohort.

Risks

Intraventricular hemorrhage

- **Severe IVH (grade III or IV) and resulting PVL** are linked to CP.

- Even **low-grade IVH** is associated with more CP at school age in extremely preterm children; higher grades carry greater risk.

Ischemia

- Immature vessels and a **pressure-passive** cerebral circulation: with hypotension the preterm brain cannot ↑ perfusion to **borderzone** areas fed by immature penetrating vessels.
- Hypoxia-ischemia → **pyramidal tract damage** → **spastic diplegia**.

Perinatal infection/inflammation

- **PVL in 9%** of 753 neonates born 24–32 wk; highest risk if born **<28 wk** and/or with inflammatory events days–weeks before delivery.
- PVL strongly associated with **prolonged membrane rupture, chorioamnionitis, neonatal hypotension**. **Chronic** (not acute) placental inflammation is linked to PVL.
- 3094 singletons <33 wk: **15%** had clinical chorioamnionitis; **IVH 22% vs 12%** (infected vs not).
- **Mechanism (Fig 34-3):**
 - Infection → **TNF, IL-1, IL-6, IL-8** → prostaglandins → **preterm labor**.
 - Same cytokines are **directly toxic to oligodendrocytes and myelin**.
 - Vessel rupture, hypoxia, cytokines → neuronal death → **glutamate** release → **Ca²⁺ influx** → white-matter toxicity (glutamate also directly toxic to oligodendrocytes).
 - Activated **microglia/macrophages** (cytokines, chemokines, proteases, complement, excitotoxins, ROS, NO) worsen secondary injury.
- TNF and IL-6 are more frequent in brains of infants who died with PVL; cytokines are linked to white-matter lesions **even without demonstrable organisms**.

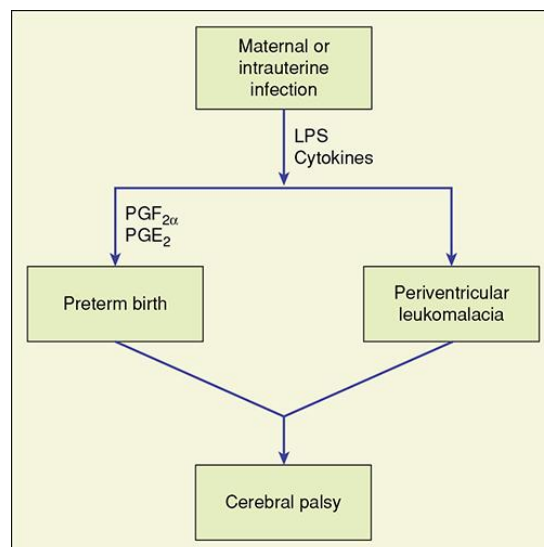


Figure 34-3. Maternal/intrauterine infection → LPS and cytokines → $\text{PGF}_{2\alpha}/\text{PGE}_2$ → preterm birth, and directly → PVL; both lead to cerebral palsy.

Prevention — neuroprotection

- **Antenatal magnesium sulfate and corticosteroids** — benefits well described.
- **Erythropoiesis-stimulating agents** (erythropoietin, darbepoetin):
 - Meta-analysis: ↓ IVH and PVL when given soon after birth; neurodevelopmental effect unclear.
 - Large trial: early high-dose **erythropoietin did not ↓ severe NDI or death at 2 yr**. Darbepoetin trial in follow-up.
- ⚠ **Therapeutic hypothermia is NOT recommended for preterm newborns** (different gray/white matter injury pattern, greater cold-stress susceptibility). Trial at **33 0/7–35 6/7 wk** with moderate/severe HIE is in follow-up.

Quick-Recall: Preterm Complications — Cause, Clue, Management

Complication	Key cause/mechanism	Classic clue	Management / prevention
RDS	Surfactant deficiency (type II pneumocytes)	Grunting, retractions; reticulogranular CXR + air bronchograms; hyaline membranes	CPAP + selective surfactant (thin catheter); antenatal steroids
BPD	Ventilation (baro/volutrauma), hyperoxia, infection	Chronic O ₂ dependence	Caffeine (≤ 1250 g), vitamin A (< 1500 g); no routine postnatal steroids or iNO
NEC	Multifactorial: immaturity, feeds, dysbiosis, hypoxia	Pneumatosis intestinalis, portal venous gas, bloody stools	Bowel rest, decompression, antibiotics, TPN; surgery for perforation
ROP	Hyperoxia → vasoconstriction → VEGF-driven neovascularization	Retinal detachment	Lower SpO ₂ targets ($\uparrow$ death trade-off); anti-VEGF
IVH	Fragile germinal matrix (involutes 34–36 wk)	Within 72 h; half silent	Antenatal steroids ≥ 48 h; delayed clamping; avoid cord milking
PVL	Ischemia, infection/cytokines	Echolucent cysts after ≥ 2 wk	Mg sulfate neuroprotection
CP (preterm)	IVH/PVL, ischemia, inflammation	Spastic diplegia	MgSO ₄ , steroids; no cooling

Key Numbers to Memorize

- Viability **22–24 wk**; survival < 29 wk **~80%**; preterm rate **10%** (2020) causes **⅓** of infant deaths
- Antenatal steroids: **24 0/7–33 6/7 wk** single course; consider from **23 wk** if delivery within **7 days**; ≥ 48 h before birth for IVH benefit; rescue course if first **>14 days** ago; late preterm **34–36 wk**
- Caffeine for **≤ 1250 g**; vitamin A for **< 1500 g**; LISA $\downarrow$ intubation in first **72 h**
- SUPPORT: SpO₂ **85–89% vs 91–95%** → death **20 vs 16%**, severe ROP **8.6 vs 17.9%**
- Retinal vascularization begins **~4th month**
- IVH: most within **72 h**; germinal matrix involutes **34–36 wk**; < 29 wk incidence **19 → 13%**; **11% at 23 wk**, **5% at 28 wk**
- Papile: I germinal matrix; II **10–50%**; III **>50%**; IV = parenchymal (venous infarction)
- PVL cysts need **≥ 2 wk** (up to **4 months**); cystic injury evolves over **2–5 wk**
- **90–96%** of preterm CP have sonographic lesions; CP **10.5%** if born < 28 wk; US CP **3/1000**
- Repeat steroids: CP RR **5.7** (NS); Australian CP **4.2 vs 4.8%**
- Chorioamnionitis < 33 wk: **15%**; IVH **22 vs 12%**; PVL **9%** at 24–32 wk